SafeStep

An AI-enabled smart insole that calls for help when you can't

Team 12 | Domain: Safety and Security | IIM Sambalpur

Avidha Chandra · Agrima · Anirudh Kumar · Yuvraj Rao Padala · Divyansh Sinha · Pratyush Phadnaik · Shreyam Sai Vinayak · Aayush Agrawal

1. INTRODUCTION

Picture travelling back late at night, phone battery at 5%, in a shared auto that has just taken a turn you don't recognise. You could call someone, but taking out your phone might not be a good idea right now.

That moment is where SafeStep started. SafeStep is a small, removable pod that fits inside an ordinary shoe insole. It has its own GPS, its own SIM, a motion sensor and a rechargeable battery. If you're in trouble, you tap your heel twice and a silent SOS goes out with your live location to three trusted contacts and your campus security desk. Nobody around you sees anything happen.

The pod also runs a small AI model that watches for things you may not be able to report yourself, such as a fall, a sudden sprint, or a stop that lasts far too long on a route you take every day. If it notices one of these, it starts an alert. You get 30 seconds to cancel it if it's a false alarm.

Alongside the shoe, we've designed **SafeRoute**, a companion app for cab and auto rides. It checks your live route against the expected one and warns you, and your contacts, if the vehicle goes noticeably off course.

2. THE PROBLEM

Safety while travelling alone is still a real and everyday concern in India. NCRB data shows 4,48,211 cases of crime against women were registered in 2023, slightly up from 4,45,256 in 2022, and more than 88,000 of these were cases of kidnapping and abduction.[1] These are only the cases that reached a police station, and the NCRB itself notes that its figures reflect recorded cases, not actual prevalence.

The fear is even more widespread than the crime. When the Ola Mobility Institute surveyed 9,935 women across 11 cities, only 9% said public transport felt "very safe".[2] Night is the worst time. A 2024 Greenpeace India report found that 77% of women felt unsafe on Delhi buses after dark.[3] Shared transport matters a great deal here, because women make up 40–45% of users of autos, on-demand taxis and other shared mobility.[2]

What struck us while researching this is that **almost every safety tool available today depends on the phone**. Helpline apps, ride-hailing SOS buttons and even smart safety jewellery all need a working phone in reach. In a real emergency, the phone is often the first thing to be snatched, switched off or lost, or it simply dies. A safety system that fails along with the phone isn't much use at the moment it's needed most.

If this gap stays open, students avoid evening classes, families limit where their daughters can go, and field workers travel to remote areas with no reliable way to call for help. When something does go wrong, every minute lost before anyone knows where the person is makes it harder to help.

3. WHAT SAFESTEP DOES

Our goal is simple: make it possible to call for help even when the phone is gone and you can't openly ask for it. The **heel-tap SOS** is the main trigger. **Automatic detection** covers moments when you can't act, and every automatic alert has a **30-second cancel window**, so a stumble on the stairs doesn't send your parents into a panic. Because the pod has its own GPS and SIM, it **doesn't need your phone**. In weak-signal areas it saves the alert and sends it as soon as the network returns.

We also took privacy seriously. SafeStep only shares location during an SOS or an active trip, and all stored data is deleted after 24 hours.

4. WHERE AI COMES IN

It helps to separate what the product does from what the AI specifically adds. The product's basic value is a **hidden emergency trigger that works without a phone**. On its own, though, that's just a panic button in a shoe. AI is what makes it reliable and proactive.

First, AI **tells a real tap apart from normal movement** like stamping, stairs or running for a bus. A small model on the pod's chip (TinyML) learns what a deliberate double tap looks like. Without this, false alarms would make the product unusable.

Second, AI **notices when something is wrong, even if you don't press anything**. The model learns your normal walking patterns and regular routes, then flags things that don't fit: a fall, a sudden sprint, or a long stop in an odd place. This matters because in many dangerous situations, the person can't trigger an alert themselves.

Third, in SafeRoute, AI **scores route deviations** instead of reacting to every wrong turn. A detour around traffic is normal, but heading the wrong way out of the city isn't. Because the motion model runs on the device itself, detection also works offline, and movement data stays on the pod unless there's an alert.

5. WHO IT'S FOR

The people we see using SafeStep are college students (especially women) travelling alone at night, NGO and healthcare field workers visiting remote areas, and working women on late shifts who rely on shared autos and cabs.

The payers won't always be the users: colleges (linked to their security desk), NGOs for field staff, companies through safety or CSR budgets, and parents.

To keep a real person in mind, we built a simple persona. **Priya** is 20 and studies at a college on the edge of the city. She usually takes a shared auto back to her hostel after evening classes, and by then her phone is nearly dead. Priya doesn't want to walk home clutching her phone, and her college wants a faster way to respond if a student is in trouble.

6. WHAT MAKES IT DIFFERENT

The demand for wearable safety devices is already proven. Leaf Wearables' Safer pendant, for example, sends an SOS when pressed twice, and General Motors India once handed the device to all its women employees.[4] But Safer connects to the user's phone over Bluetooth, so it shares the same weak point we described above.

Capability	Phone apps / ride-app SOS	Smart pendants (e.g. Safer)	SafeStep
Works if the phone is taken or dead	No	No	**Yes**
Can be triggered without anyone noticing	No	Partly	**Yes**
Raises an alert without user action	No	No	**Yes**
Monitors the trip route	Some	No	**Yes**

Table 1. SafeStep compared with existing personal-safety solutions

SafeStep improves on existing safety wearables. It is **more reliable** because it works without a phone, **more discreet** because it's hidden inside your shoe, and **more proactive** because the AI can raise an alarm on its own.

UNIQUE VALUE PROPOSITION

SafeStep calls for help even when you can't.

7. OTHER USES

Campus safety is our starting point, but the same pod fits several other settings: ASHA workers and survey teams in rural areas, elderly care (through fall detection), children's routes to school, trekking in low-network areas, and companies running night-shift transport.

8. PROTOTYPE AND FEASIBILITY

Functional prototype. To show SafeStep working, we built a web app in which a smartphone stands in for the pod. It uses the phone's real motion sensor to detect a double-tap or a fall, runs the 30-second cancel window, and sends an SOS with GPS location to a live campus security dashboard, which can reply that a guard is on the way. It also replays a SafeRoute trip and calculates how far the ride strays from its planned route.

Physical product. Everything in the pod is off-the-shelf. For the hardware prototype we'd use an ESP32 microcontroller, a NEO-6M GPS module, a SIM800L GSM module (to be replaced by a 4G LTE-M module in production), an MPU6050 motion sensor and a small Li-Po battery, all in a waterproof casing. The app would be built in Flutter on top of the Google Maps API. We'd train the motion model with TensorFlow Lite, using data we record ourselves while walking, running and tapping.

Our rough estimate is about ₹1,500 per unit for the prototype parts, with a target retail price of ₹2,499. A monthly subscription of around ₹99 would cover the SIM data and the monitoring service. Colleges and NGOs would buy in bulk, and CSR funding could subsidise devices for rural women and field workers.

Battery life is the biggest risk, so the pod uses low-power sleep modes and can be removed for charging. False alarms are handled by the AI tap detection and the cancel window. Our plan is to build and test the prototype in three months, run a pilot with about 100 students and one NGO field team in months three to six, and then look for a footwear brand partner and CSR funding to scale.

9. CONCLUSION

Most safety tools assume you'll have your phone in hand and the freedom to use it. SafeStep is built for the moments when you don't. A phone-independent device in the shoe, with AI that recognises danger, offers protection that works when it's needed, and affordable parts and clear institutional buyers make it realistic to build and scale.

REFERENCES

1. National Crime Records Bureau, *Crime in India* 2023 (released September 2025), as reported by ETV Bharat and Morung Express.
2. Ola Mobility Institute, *What Do Women and Girls Want from Urban Mobility Systems?* (2019), as reported by Business Standard and Zee Business.
3. Greenpeace India, *Riding the Justice Route* (2024), as reported by Business Standard.
4. Leaf Wearables (Safer): YourStory (2015); Pitstop Weekly.

Cost and pricing figures in Section 8 are the team's preliminary estimates.